

Differential Equations Notes: Second-Order Linear Homogeneous & Non-Homogeneous ODEs

Hemanta Mandal

1 Second-Order Homogeneous Linear DEs (IMG_1006)

Standard Form of 2nd Order Linear DE:

$$y'' + P(x)y' + Q(x)y = r(x)$$

Examples of Linear and Non-Linear DEs

- $x^2y'' + 2xy' + 6y = 0 \implies$ Linear Homogeneous
 - $y'' + y'^2 + 2y = e^{-x} \cos x \implies$ Non-Homogeneous, Non-Linear
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Example 1: Fundamental Solutions

Solve:

$$y'' + y = 0$$

Two linearly independent solutions are:

$$y_1 = \cos x, \quad y_2 = \sin x$$

Verification:

$$y_1' = -\sin x, \quad y_1'' = -\cos x \implies y_1'' + y_1 = -\cos x + \cos x = 0$$

$$y_2' = \cos x, \quad y_2'' = -\sin x \implies y_2'' + y_2 = -\sin x + \sin x = 0$$

So the general solution is:

$$\mathbf{y_g(x)} = \mathbf{c_1y_1} + \mathbf{c_2y_2} = \mathbf{c_1 \cos x} + \mathbf{c_2 \sin x}$$

Superposition Principle (Theorem)

Theorem: If y_1 and y_2 are two linearly independent solutions of the homogeneous linear DE $y'' + P(x)y' + Q(x)y = 0$, then:

$$y = c_1y_1 + c_2y_2 \quad \text{is also a solution.}$$

Proof: Since y_1 and y_2 are solutions:

$$y_1'' + Py_1' + Qy_1 = 0 \quad \text{--- (1)}$$

$$y_2'' + Py_2' + Qy_2 = 0 \quad \text{--- (2)}$$

Let $y = c_1y_1 + c_2y_2$, then $y' = c_1y_1' + c_2y_2'$ and $y'' = c_1y_1'' + c_2y_2''$.

Substituting into the DE (IMG_1007):

$$\begin{aligned} & (c_1y_1'' + c_2y_2'') + P(c_1y_1' + c_2y_2') + Q(c_1y_1 + c_2y_2) \\ &= c_1(y_1'' + Py_1' + Qy_1) + c_2(y_2'' + Py_2' + Qy_2) \end{aligned}$$

Using (1) and (2):

$$= c_1(0) + c_2(0) = 0$$

Hence $y = c_1y_1 + c_2y_2$ satisfies the equation and is a general solution.

2 Reduction of Order Method (IMG_1007 – IMG_1009)

Finding a second solution y_2 given one known solution y_1 of $y'' + P(x)y' + Q(x)y = 0$.

Let $y_2 = u(x)y_1$, where $u(x)$ is an unknown function.

$$y_2' = u'y_1 + uy_1'$$

$$y_2'' = u''y_1 + 2u'y_1' + uy_1''$$

Since y_2 is a solution:

$$(u''y_1 + 2u'y_1' + uy_1'') + P(u'y_1 + uy_1') + Q(uy_1) = 0$$

$$u''y_1 + u'(2y_1' + Py_1) + u(y_1'' + Py_1' + Qy_1) = 0$$

Since y_1 is a solution, the term in the last bracket is 0:

$$u''y_1 + u'(2y_1' + Py_1) = 0$$

Divide by y_1 :

$$u'' + u' \left(\frac{2y_1'}{y_1} + P \right) = 0$$

Let $U = u' \implies U' = u''$:

$$U' + U \left(\frac{2y_1'}{y_1} + P \right) = 0 \implies \frac{dU}{U} = - \left(\frac{2y_1'}{y_1} + P \right) dx$$

Integrating both sides:

$$\ln U = -2 \ln y_1 - \int P dx \implies U = y_1^{-2} e^{-\int P dx}$$

Since $u' = U$, we integrate to find u :

$$u = \int U dx = \int y_1^{-2} e^{-\int P dx} dx$$

Thus, the second independent solution is:

$$y_2 = y_1 \int y_1^{-2} e^{-\int P dx} dx$$

Example 1: Reduction of Order

Given $y_1 = x$ is a solution of $x^2 y'' - xy' + y = 0$, find the second solution.

Step 1: Convert to Standard Form

$$y'' - \frac{1}{x}y' + \frac{1}{x^2}y = 0 \implies P(x) = -\frac{1}{x}, \quad Q(x) = \frac{1}{x^2}$$

Step 2: Calculate U

$$U = y_1^{-2} e^{-\int P dx} = \frac{1}{x^2} e^{-\int -\frac{1}{x} dx} = \frac{1}{x^2} e^{\ln x} = \frac{x}{x^2} = \frac{1}{x}$$

Step 3: Calculate u and y_2

$$u = \int U dx = \int \frac{1}{x} dx = \ln x$$

$$y_2 = y_1 u = x \ln x$$

3 Second-Order Linear Homogeneous DE with Constant Coefficients (IMG_1010 – IMG_1011)

Standard Form:

$$y'' + ay' + by = 0$$

Assume trial solution $y = e^{\lambda x} \implies y' = \lambda e^{\lambda x}, \quad y'' = \lambda^2 e^{\lambda x}$.

$$\lambda^2 e^{\lambda x} + a\lambda e^{\lambda x} + b e^{\lambda x} = 0 \implies e^{\lambda x}(\lambda^2 + a\lambda + b) = 0$$

Since $e^{\lambda x} \neq 0$, we get the **Characteristic Equation**:

$$\lambda^2 + a\lambda + b = 0$$

Three Cases of Roots (λ_1, λ_2)

Case 1: Real and Distinct Roots ($\lambda_1 \neq \lambda_2$)

$$y = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

Example (IMG_1010): Solve $y'' - y = 0 \implies \lambda^2 - 1 = 0 \implies \lambda = \pm 1$.

$$y = c_1 e^x + c_2 e^{-x}$$

Case 2: Real and Equal Roots ($\lambda_1 = \lambda_2 = \lambda$)

Here $\lambda = -\frac{a}{2}$. Using reduction of order:

$$y_1 = e^{-\frac{a}{2}x}, \quad y_2 = xe^{-\frac{a}{2}x}$$

$$\mathbf{y} = (\mathbf{c}_1 + \mathbf{c}_2 \mathbf{x})\mathbf{e}^{\lambda \mathbf{x}}$$

Example (IMG_1011): Solve $y'' + 8y' + 16y = 0 \implies (\lambda + 4)^2 = 0 \implies \lambda_1 = \lambda_2 = -4$.

$$\mathbf{y} = (\mathbf{c}_1 + \mathbf{c}_2 \mathbf{x})\mathbf{e}^{-4\mathbf{x}}$$

Case 3: Complex Conjugate Roots ($\lambda = a \pm bi$)

Solutions in terms of real functions:

$$\mathbf{y} = \mathbf{e}^{a\mathbf{x}}(\mathbf{c}_1 \cos b\mathbf{x} + \mathbf{c}_2 \sin b\mathbf{x})$$

Example (IMG_1011): Solve $y'' - 2y' + 2y = 0 \implies \lambda^2 - 2\lambda + 2 = 0 \implies \lambda = \frac{2 \pm \sqrt{4-8}}{2} = 1 \pm i$.

$$\text{Here } a = 1, b = 1 \implies \mathbf{y} = \mathbf{e}^{\mathbf{x}}(\mathbf{c}_1 \cos \mathbf{x} + \mathbf{c}_2 \sin \mathbf{x})$$

4 Second-Order Non-Homogeneous Linear DE (IMG_1012 – IMG_1015)

Standard Form:

$$y'' + P(x)y' + Q(x)y = r(x)$$

General Solution:

$$\mathbf{y}(\mathbf{x}) = \mathbf{y}_h(\mathbf{x}) + \mathbf{y}_p(\mathbf{x})$$

where $y_h(x) = c_1 y_1(x) + c_2 y_2(x)$ is the complementary/homogeneous solution, and $y_p(x)$ is a particular solution.

Methods to find y_p :

1. Method of Undetermined Coefficients
 2. Variation of Parameters
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Method of Undetermined Coefficients Choice Table (IMG_1012)

Rules:

1. Choose initial trial form y_p from table.
 2. If any term in y_p is already present in y_h , multiply y_p by x (or x^2).
 3. If $r(x)$ is a sum of functions, y_p is the sum of respective trial forms.
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Right-Hand Side $r(x)$	Trial Choice for y_p
ke^{ax}	Ce^{ax}
$a_n x^n + \cdots + a_1 x + a_0$	$C_n x^n + \cdots + C_1 x + C_0$
$k \cos ax$ or $k \sin ax$	$C_1 \cos ax + C_2 \sin ax$
$ke^{ax} \cos bx$ or $ke^{ax} \sin bx$	$e^{ax}(A \cos bx + B \sin bx)$

Example 1 (IMG_1013)

Solve:

$$y'' - 3y' + 2y = e^{3x}$$

Step 1: Homogeneous Solution y_h

$$\lambda^2 - 3\lambda + 2 = 0 \implies (\lambda - 1)(\lambda - 2) = 0 \implies \lambda = 1, 2$$

$$y_h = c_1 e^x + c_2 e^{2x}$$

Step 2: Particular Solution y_p Try $y_p = Ce^{3x} \implies y'_p = 3Ce^{3x}, \quad y''_p = 9Ce^{3x}$.
Substitute into DE:

$$9Ce^{3x} - 3(3Ce^{3x}) + 2(Ce^{3x}) = e^{3x}$$

$$(9C - 9C + 2C)e^{3x} = e^{3x} \implies 2C = 1 \implies C = \frac{1}{2}$$

$$y_p = \frac{1}{2}e^{3x}$$

General Solution:

$$y = c_1 e^x + c_2 e^{2x} + \frac{1}{2}e^{3x}$$

Example 2 (IMG_1014)

Solve:

$$y'' - 3y' + 2y = 3 \sin 2x$$

Homogeneous solution: $y_h = c_1 e^x + c_2 e^{2x}$.

Trial Solution for y_p :

$$y_p = A \cos 2x + B \sin 2x$$

$$y'_p = -2A \sin 2x + 2B \cos 2x$$

$$y''_p = -4A \cos 2x - 4B \sin 2x$$

Substitution:

$$(-4A \cos 2x - 4B \sin 2x) - 3(-2A \sin 2x + 2B \cos 2x) + 2(A \cos 2x + B \sin 2x) = 3 \sin 2x$$

Grouping terms:

$$\cos 2x(-4A - 6B + 2A) + \sin 2x(-4B + 6A + 2B) = 3 \sin 2x$$

$$\cos 2x(-2A - 6B) + \sin 2x(6A - 2B) = 3 \sin 2x$$

Equating coefficients:

$$\begin{cases} -2A - 6B = 0 & \implies A = -3B \\ 6A - 2B = 3 \end{cases}$$

Substitute $A = -3B$:

$$6(-3B) - 2B = 3 \implies -20B = 3 \implies B = -\frac{3}{20}$$

$$A = -3 \left(-\frac{3}{20} \right) = \frac{9}{20}$$

General Solution:

$$y = c_1 e^x + c_2 e^{2x} + \frac{9}{20} \cos 2x - \frac{3}{20} \sin 2x$$

Proof of Non-Homogeneous General Solution Structure (IMG_1015)

Let $y(x) = c_1 y_1(x) + c_2 y_2(x) + y_p(x)$. Substituting into $a_2(x)y'' + a_1(x)y' + a_0(x)y$:

$$\begin{aligned} &= a_2(x)(c_1 y_1 + c_2 y_2 + y_p)'' + a_1(x)(c_1 y_1 + c_2 y_2 + y_p)' + a_0(x)(c_1 y_1 + c_2 y_2 + y_p) \\ &= c_1[a_2 y_1'' + a_1 y_1' + a_0 y_1] + c_2[a_2 y_2'' + a_1 y_2' + a_0 y_2] + [a_2 y_p'' + a_1 y_p' + a_0 y_p] \\ &= c_1(0) + c_2(0) + r(x) = r(x) \end{aligned}$$

This proves that $y = y_h + y_p$ satisfies the non-homogeneous equation.

5 Method of Variation of Parameters (IMG_1016)

Used when the method of undetermined coefficients is non-applicable or inconvenient.

Given a second-order linear non-homogeneous differential equation:

$$y'' + P(x)y' + Q(x)y = r(x)$$

The particular solution is given by:

$$y_p(x) = -y_1 \int \frac{y_2 \cdot r(x)}{W} dx + y_2 \int \frac{y_1 \cdot r(x)}{W} dx$$

where $y_1(x)$ and $y_2(x)$ form a fundamental set of solutions (basis) for the corresponding homogeneous equation:

$$y'' + P(x)y' + Q(x)y = 0$$

and $W = W(y_1, y_2)$ is the ****Wronskian**** of y_1 and y_2 :

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = y_1 y_2' - y_2 y_1'$$

Example 1 (IMG_1016 & IMG_1017)

Solve:

$$y'' + 9y = \sec 3x$$

Step 1: Homogeneous Solution (y_h)

$$\lambda^2 + 9 = 0 \implies \lambda = \pm 3i$$

$$y_h = c_1 \cos 3x + c_2 \sin 3x$$

Here, $y_1 = \cos 3x$, $y_2 = \sin 3x$, and $r(x) = \sec 3x$.

Step 2: Wronskian (W)

$$W = \begin{vmatrix} \cos 3x & \sin 3x \\ -3 \sin 3x & 3 \cos 3x \end{vmatrix} = 3 \cos^2 3x + 3 \sin^2 3x = 3$$

Step 3: Particular Solution (y_p)

$$\begin{aligned} y_p &= -y_1 \int \frac{y_2 \cdot r(x)}{W} dx + y_2 \int \frac{y_1 \cdot r(x)}{W} dx \\ y_p &= -\cos 3x \int \frac{\sin 3x \cdot \sec 3x}{3} dx + \sin 3x \int \frac{\cos 3x \cdot \sec 3x}{3} dx \\ &= -\cos 3x \int \frac{\tan 3x}{3} dx + \sin 3x \int \frac{1}{3} dx \\ &= -\cos 3x \cdot \frac{1}{9} (-\ln |\cos 3x|) + \frac{1}{3} x \sin 3x \\ &= \frac{1}{9} \cos 3x \ln |\cos 3x| + \frac{1}{3} x \sin 3x \end{aligned}$$

General Solution:

$$y = y_h + y_p = c_1 \cos 3x + c_2 \sin 3x + \frac{1}{9} \cos 3x \ln |\cos 3x| + \frac{1}{3} x \sin 3x$$

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Example 2 (IMG_1018)

Solve:

$$y'' - 4y' + 4y = \frac{e^{2x}}{x}$$

Step 1: Homogeneous Solution (y_h)

$$\lambda^2 - 4\lambda + 4 = 0 \implies (\lambda - 2)^2 = 0 \implies \lambda = 2, 2$$

$$y_h = c_1 e^{2x} + c_2 x e^{2x}$$

Here, $y_1 = e^{2x}$, $y_2 = x e^{2x}$, and $r(x) = \frac{e^{2x}}{x}$.

Step 2: Wronskian (W)

$$W = \begin{vmatrix} e^{2x} & xe^{2x} \\ 2e^{2x} & e^{2x} + 2xe^{2x} \end{vmatrix} = e^{4x} + 2xe^{4x} - 2xe^{4x} = e^{4x}$$

Step 3: Particular Solution (y_p)

$$\begin{aligned} y_p &= -e^{2x} \int \frac{xe^{2x} \cdot \left(\frac{e^{2x}}{x}\right)}{e^{4x}} dx + xe^{2x} \int \frac{e^{2x} \cdot \left(\frac{e^{2x}}{x}\right)}{e^{4x}} dx \\ &= -e^{2x} \int 1 dx + xe^{2x} \int \frac{1}{x} dx \\ &= -xe^{2x} + xe^{2x} \ln x \end{aligned}$$

General Solution:

$$y = y_h + y_p = (c_1 + c_2 x)e^{2x} - xe^{2x} + xe^{2x} \ln x$$

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6 Finding Particular Integrals by Inverse Operator Method (IMG_1019 – IMG_1022)

Let $f(D)y = r(x)$, then $y_p = \frac{1}{f(D)}r(x)$.

Case 1: Exponential Functions $r(x) = e^{ax}$ (IMG_1019)

$$y_p(x) = \frac{1}{f(D)}e^{ax} = \frac{1}{f(a)}e^{ax}, \quad \text{if } f(a) \neq 0$$

If $f(a) = 0$, then:

$$y_p(x) = x \frac{1}{f'(a)}e^{ax}$$

If $f'(a) = 0$, then:

$$y_p(x) = x^2 \frac{1}{f''(a)}e^{ax}, \quad \text{and so on.}$$

Example (IMG_1019): Find y_p for $(D^2 - 3D + 2)y = e^{3x}$.

$$y_p = \frac{1}{D^2 - 3D + 2}e^{3x} = \frac{1}{3^2 - 3(3) + 2}e^{3x} = \frac{1}{2}e^{3x}$$

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Case 2: Sine and Cosine Functions $r(x) = \sin(ax + b)$ **or** $\cos(ax + b)$ (IMG_1020)

$$P.I. = \frac{1}{f(D^2)} \sin(ax + b) = \frac{1}{f(-a^2)} \sin(ax + b), \quad \text{if } f(-a^2) \neq 0$$

If $f(-a^2) = 0$, then:

$$P.I. = x \frac{1}{f'(-a^2)} \sin(ax + b)$$

(A similar result holds for $\cos(ax + b)$).

Example (IMG_1020): Find $P.I.$ for $y'' + 4y' + y = \cos(2x - 1) \implies (D^2 + 4D + 1)y = \cos(2x - 1)$. Here $a = 2 \implies D^2 \rightarrow -2^2 = -4$.

$$P.I. = \frac{1}{D^2 + 4D + 1} \cos(2x - 1) = \frac{1}{-4 + 4D + 1} \cos(2x - 1) = \frac{1}{4D - 3} \cos(2x - 1)$$

Multiply numerator and denominator by $(4D + 3)$ or $(1 + 4D)$:

$$= \frac{1 + 4D}{(1 - 4D)(1 + 4D)} \cos(2x - 1) = \frac{1 + 4D}{1 - 16D^2} \cos(2x - 1)$$

Substitute $D^2 \rightarrow -4$:

$$\begin{aligned} &= \frac{(1 + 4D) \cos(2x - 1)}{1 - 16(-4)} = \frac{1}{65} (\cos(2x - 1) + 4D[\cos(2x - 1)]) \\ &= \frac{1}{65} (\cos(2x - 1) - 8 \sin(2x - 1)) \end{aligned}$$

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Case 3: Polynomial Functions $r(x) = x^m$ (IMG_1021)

Express $f(D)$ in terms of $[1 + \phi(D)]$ and expand using Binomial expansion $(1 + x)^{-1} = 1 - x + x^2 - x^3 + \dots$

Example (IMG_1021): Solve $\frac{d^2y}{dx^2} + \frac{dy}{dx} = x^2 + 2x + 4 \implies (D^2 + D)y = x^2 + 2x + 4$.

$$\begin{aligned} y_p &= \frac{1}{D^2 + D} (x^2 + 2x + 4) = \frac{1}{D(1 + D)} (x^2 + 2x + 4) \\ &= \frac{1}{D} (1 + D)^{-1} (x^2 + 2x + 4) = \frac{1}{D} [1 - D + D^2 - D^3 + \dots] (x^2 + 2x + 4) \end{aligned}$$

Applying derivatives:

$$\begin{aligned} 1(x^2 + 2x + 4) &= x^2 + 2x + 4 \\ -D(x^2 + 2x + 4) &= -(2x + 2) \\ D^2(x^2 + 2x + 4) &= 2 \end{aligned}$$

Summing up terms:

$$= \frac{1}{D} [(x^2 + 2x + 4) - (2x + 2) + 2] = \frac{1}{D} [x^2 + 4]$$

Integrating with respect to x :

$$= \int (x^2 + 4) dx = \frac{x^3}{3} + 4x$$

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Case 4: Exponential Shift $r(x) = e^{ax}V(x)$ (IMG_1021 & IMG_1022)

$$\frac{1}{f(D)}[e^{ax}V(x)] = e^{ax}\frac{1}{f(D+a)}V(x)$$

Example (IMG_1022): Find $P.I.$ for $(D^2 - 2D + 4)y = e^x \cos x$.

Here $a = 1$ and $V(x) = \cos x$.

$$\begin{aligned} P.I. &= \frac{1}{D^2 - 2D + 4}(e^x \cos x) = e^x \frac{1}{(D+1)^2 - 2(D+1) + 4} \cos x \\ &= e^x \frac{1}{(D^2 + 2D + 1) - 2D - 2 + 4} \cos x = e^x \frac{1}{D^2 + 3} \cos x \end{aligned}$$

For $\cos x$, $a = 1 \implies D^2 \rightarrow -1^2 = -1$:

$$= e^x \frac{1}{-1 + 3} \cos x = \frac{1}{2} e^x \cos x$$